Disclaimer: This document is meant for you to train for the final exam and check your knowledge about the topics in the MDK course. It is NOT meant to be (and will NOT be) a "fac simile" of the exam, since you will have to proof that you mastered the topics at a level in which you can apply the knowledge you gained to new problems and new questions. I suggest to try to solve these trainings after self study and confrontation with collegues, to assess your level of preparation.

1 Multiple Choice Questions

Read carefully. There is no limit in the number of right or wrong answers.

1. Since $so(3)$ is a Lie algebra, it possesses an antisymmetric operator called the commutator given by $[\tilde{\omega}_1, \tilde{\omega}_2] := \tilde{\omega}_1\tilde{\omega}_2 - \tilde{\omega}_2\tilde{\omega}_1; \quad \tilde{\omega}_1, \tilde{\omega}_2 \in so(3)$. What is true about the commutator?
   - ◯ $[\tilde{\omega}_1, \tilde{\omega}_2] \in so(3)$
   - ◯ $[\tilde{\omega}_1, \tilde{\omega}_2] \in so^*(3)$
   - ◯ $[\tilde{\omega}_1, \tilde{\omega}_2] = 0$ implies that either $\omega_1 = 0$ or $\omega_2 = 0$
   - ◯ $[\tilde{\omega}, \tilde{\omega}] = 0$ if and only if $\tilde{\omega} = 0$
   - ◯ The commutator of its arguments in tilde form represents the scalar product of its arguments in vector form.
   - ◯ $[\tilde{\omega}_1, \tilde{\omega}_2] = 0$ implies that $\tilde{\omega}_1$ and $\tilde{\omega}_2$ commute.

2. Consider a rotating rigid body that is pinned from its center. Attached to the rotating body is the body-fixed frame Ψ_B. Now consider another fixed-intertial frame Ψ_I with the same origin as Ψ_B. The configuration of the body is described by R_B^I. Let p be a point attached to the rotating body. What is the velocity of p expressed in Ψ_I?
   - ◯ $\dot{p}^i = \omega_I^{B,B} \wedge p^i$
   - ◯ $\dot{p}^i = \omega_I^{B,I} \wedge p^i$
   - ◯ $\dot{p}^i = \omega_B^{I,I} \wedge p^i$
   - ◯ $\dot{p}^i = \omega_B^{B,I} \wedge p^i$

3. Consider the Lie group $SE(3)$. What is true about the space $T_H^*SE(3)$, that is the cotangent space on $SE(3)$ at the configuration $H \in SE(3)$?

- ◯ There is a duality pairing between a twist $T \in se(3)$ and an element of $T_H^*SE(3)$, and the resulting scalar represents mechanical power along a rigid body motion.
- ◯ It is a 6-dimensional vector space.
- ◯ It has a natural inner product.
- ◯ An element of $T_H^*SE(3)$ can be naturally paired with the vector $\dot{H}$ and the resulting scalar represents mechanical power along a rigid body motion.
- ◯ It can be made into a Lie Group with a proper change of coordinates.
- ◯ It is the dual space of the space $T_H SE(3)$.

1. The matrix exponential is a map from $so(3)$ to $SO(3)$. If we want to obtain $R_A^B(t) = e^{Xt}R_A^B(0)$ what should be in the place of X?
   - ◯ $\tilde{\omega}_A^{B,B}$
   - ◯ $T_A^{A,A}$
   - ◯ $\tilde{T}_A^{B,A}$
   - ◯ $\omega_B^{A,B}$
   - ◯ W^b
   - ◯ $\dot{H}_A^B$

2. The Euler-Lagrange equations in Robotics:
   - ◯ are an alternative way to derive the equations with respect to the least action principle.
   - ◯ are equations on the space of generalised forces on the configuration manifold T^*Q.
   - ◯ are equations of the space of generalised velocities on the configuration manifold TQ
   - ◯ Are n, with n the dimension of the configuration manifold Q, i.e. the DoFs of the Robot.

3. The Geometric Jacobian:
   - ◯ Does not depend on the configuration of the robot.

- ◯ Contains the expression of the unit twists which characterise the kinematic pairs of the robot, expressed in the base frame Ψ_0.
- ◯ has always a non null determinant if the robot is redundant.
- ◯ May loose rank in some configurations.
- ◯ Is a $(0-2)$ tensor.
- ◯ Represents a linear transformation from TQ to $se(3)$

4. The inertia tensor $\mathcal{I}$ of a rigid body:
   - ◯ Is a linear transformation from $se(3)$ to $se(3)$
   - ◯ Is a $(0-2)$ tensor.
   - ◯ Is constant only in frames which are fixed with the rigid body.
   - ◯ Is diagonal in any body fixed frame.
   - ◯ Is constant only in the Principal Inertia Frame.
   - ◯ In case the rigid body reduces to a point it just represents the mass of the resulting point mass.

2 Part 2: Open Question (weight=3)

1) Consider the dynamic equation (11) of a manipulator not touching the environment and ideally without friction, with mass matrix $M(q)$, $C(q,\dot{q})$ modelling the Coriolis and centrifugal forces, and $G(q)$ the gravitational forces due to the potential energy $V(q)$ and τ are the torques applied on the joints:

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = \tau \quad , \quad G(q) := \frac{\partial V}{\partial q}(q) \tag{11}$$

How would the equation (11) change if a linear viscous friction with constant damping coefficient $b_i > 0$ is present at joint i, with $1 \leq i \leq n$? And how would it change in case an external wrench W is applied to the end effector, and expressed numerically in the end-effector frame as W^n? Motivate your answer.

2) Describe the relation between a change of coordinates between two frames (which we normally represent through an homogeneous matrix) and an isometric motion on the 3-D Euclidean space.

3) A manipulator is called redundant when it has $n > 6$ degrees of freedom. What is the value of the determinant of the geometric Jacobian matrix for such manipulators?

3 Part 3: Exercise

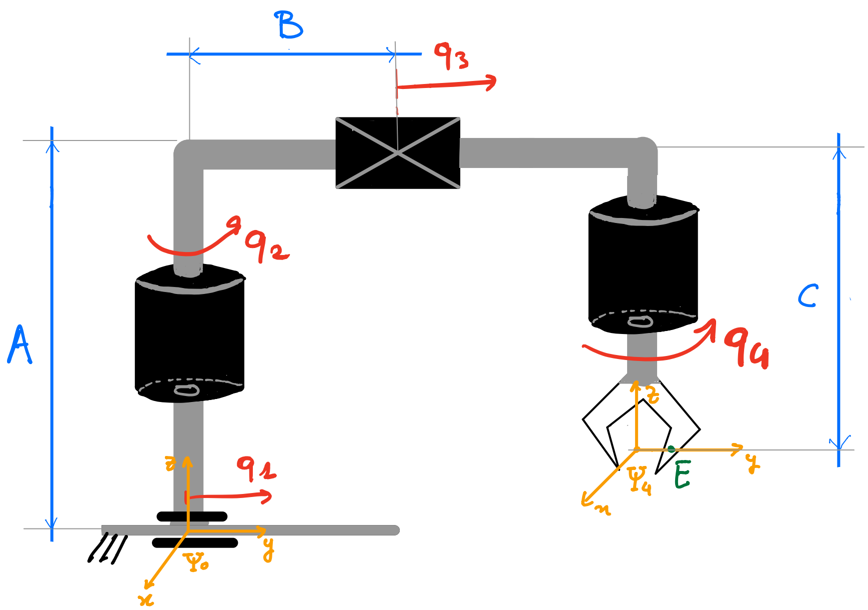
Consider the 4-DoF robot, the frames and the distances represented in the figure. The robot consists of:

- A translational joint (q_1) with axis aligned in direction y of the base frame Ψ_0;
- A rotational joint (q_2) as in figure;
- A translational joint (q_3) allowing an extension of the horizontal arm;
- A rotational joint (q_4) as in figure.

The point E is part of the end effector and has constant coordinates $(0, 1, 0)^T$ in Ψ_4, i.e. has homogeneous coordinates $E^4 = (0, 1, 0, 1)^T$ in Ψ_4. Consider as reference configuration for the first prismatic joint and the rotational joints ($q_1 = q_2 = q_4 = 0$) the one represented in the figure, and make a choice for the reference configuration of the prismatic joint (i.e. the configuration corresponding to $q_3 = 0$), considering that the distance B in the y direction is a fixed constant which cannot be shrunk (i.e. the horizontal arm has length $B + q_3$ with $q_3 \geq 0$).

1. Give Brockett's equation for direct kinematics and all unit twists in reference configurations that you need to calculate $H_4^0(q)$. *Note: There is no need to write out the matrix exponentials but you must calculate the vector products that appear inside your expressions.*
2. Compute the geometric Jacobian $J(q)$ such that $T_4^{0,0} = J(q)\dot{q}$. *Note: You must calculate the vector products that appear inside your expressions.*
3. Compute the velocity of the point E in the inertial frame Ψ_0 as function of q and $\dot{q}$.
4. Suppose an external wrench is applied on the end-effector with coordinates in Ψ_4 given by W^4=(0 0 0 0 1 0), i.e. the wrench is constant in the end-effector frame. This could be the case if e.g. a propeller mounted of

the end-effector would produce a constant thrust. Compute the generalised forces, as function of q and $\dot{q}$, which are experienced by the joints as effect caused by this wrench.



B
q3
A
q2
C
q4
z
q1
y
Ψ0
x
z
y
E
Ψ4
x